

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-0923

Snapper Creek 138 kV
December 24, 2018

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0923. The Interconnection Customer proposes a 317.6 MW solar photovoltaic facility interconnecting at the Snapper Creek 138 kV substation in the Pennsylvania area. The request was received on January 18, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 406 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 13 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$38,447,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Pulpgen - Northampton Generating Company Lp 138.0 kV
Contingency: Saint Hedwig - Snapper Creek Substation 138.0 kV L
Loading: 183.8% (Rating: 183 MVA)
DFAX: 52.5%
Pre-project loading: 92.7%
2. Facility: St Hedwig - Snapper Creek 138.0 kV
Contingency: Pulpgen - Northampton Generating Company Lp 138.0 kV line
Loading: 180.8% (Rating: 244 MVA)
DFAX: 84.2%
Pre-project loading: 71.1%
3. Facility: St Hedwig - Steel City 138.0 kV
Contingency: Northampton Generating Company Lp - Pulpgen 138.0 kV
Loading: 158.7% (Rating: 264 MVA)
DFAX: 84.2%
Pre-project loading: 57.2%
4. Facility: Northampton Generating Company Lp - Hogal 138.0 kV
Contingency: St Hedwig - Snapper Creek 138.0 kV L
Loading: 147.4% (Rating: 215 MVA)
DFAX: 52.5%
Pre-project loading: 69.8%
5. Facility: Pulpgen - Covelo 138.0 kV
Contingency: St Hedwig - Snapper Creek 138.0 kV L
Loading: 146.0% (Rating: 176 MVA)
DFAX: 100.0%
Pre-project loading: 34.6%

6. Facility: San German 6403 - Hogal 138.0 kV Line
Contingency: Saint Hedwig - Snapper Ck 138.0 kV Line
Loading: 133.4% (Rating: 229 MVA)
DFAX: 52.5%
Pre-project loading: 60.5%
7. Facility: Covelo - Scitico 138.0 kV
Contingency: Snapper Crk - St Hedwig 138.0 kV
Loading: 132.5% (Rating: 218 MVA)
DFAX: 100.0%
Pre-project loading: 12.9%
8. Facility: Scitico - Snapper Crk 138.0 kV Line
Contingency: St Hedwig - Snapper Crk 138.0 kV
Loading: 130.4% (Rating: 231 MVA)
DFAX: 100.0%
Pre-project loading: 6.9%
9. Facility: San German 6403 - Holly Springs 138.0 kV
Contingency: Saint Hedwig - Snapper Crk 138.0 kV L
Loading: 125.1% (Rating: 235 MVA)
DFAX: 52.5%
Pre-project loading: 54.1%
10. Facility: Apollo - Holly Springs 138.0 kV
Contingency: St Hedwig - Snapper Creek Substation 138.0 kV
Loading: 113.7% (Rating: 248 MVA)
DFAX: 52.5%
Pre-project loading: 46.5%
11. Facility: Guanica Tc - Rialto 138.0 kV line
Contingency: Steel City 345-138 kV Autotransformer
Loading: 109.8% (Rating: 173 MVA)
DFAX: 35.8%
Pre-project loading: 44.0%
12. Facility: Seven Springs - Apollo 138.0 kV
Contingency: St Hedwig - Snapper Creek 138.0 kV
Loading: 105.2% (Rating: 259 MVA)
DFAX: 52.5%
Pre-project loading: 40.7%
13. Facility: Steel City - Guanica Tc 138.0 kV Line
Contingency: Steel City 345/138 kV Transformer Bank
Loading: 102.1% (Rating: 234 MVA)
DFAX: 35.8%
Pre-project loading: 53.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Pulpgen - Northampton Generating Company Lp 138.0 kV	Rebuild line with higher-capacity conductor	\$3,808,000	100% (\$3,808,000)
Saint Hedwig - Snapper Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$1,454,000	100% (\$1,454,000)
Saint Hedwg - Steel City 138.0 kV	Rebuild line with higher-capacity conductor	\$895,000	100% (\$895,000)
Northampton Generating Company Lp - Hogal 138.0 kV	Rebuild line with higher-capacity conductor	\$2,721,000	100% (\$2,721,000)
Pulpgen - Covelo 138.0 kV	Rebuild line with higher-capacity conductor	\$5,467,000	100% (\$5,467,000)
San German 6403 - Hogal 138.0 kV	Rebuild line with higher-capacity conductor	\$3,533,000	100% (\$3,533,000)
Covelo - Scitico 138.0 kV	Rebuild line with higher-capacity conductor	\$1,801,000	100% (\$1,801,000)
Scitico - Snapper Creek 138.0 kV	Rebuild line with higher-capacity conductor	\$5,449,000	100% (\$5,449,000)
San German 6403 - Holly Springs 138.0 kV	Rebuild line with higher-capacity conductor	\$7,116,000	100% (\$7,116,000)
Apollo - Holly Springs 138.0 kV	Rebuild line with higher-capacity conductor	\$3,280,000	100% (\$3,280,000)
Guanica Tc - Rialto 138.0 kV line	Reconductor line	\$660,000	80% (\$530,000)
Seven Springs - Apollo 138.0 kV	Reconductor line	\$1,223,000	100% (\$1,223,000)
Steel City - Guanica Tc 138.0 kV Line	Reconductor line	\$1,457,000	80% (\$1,170,000)
	Total allocated to project		\$38,447,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.